Rack & Power

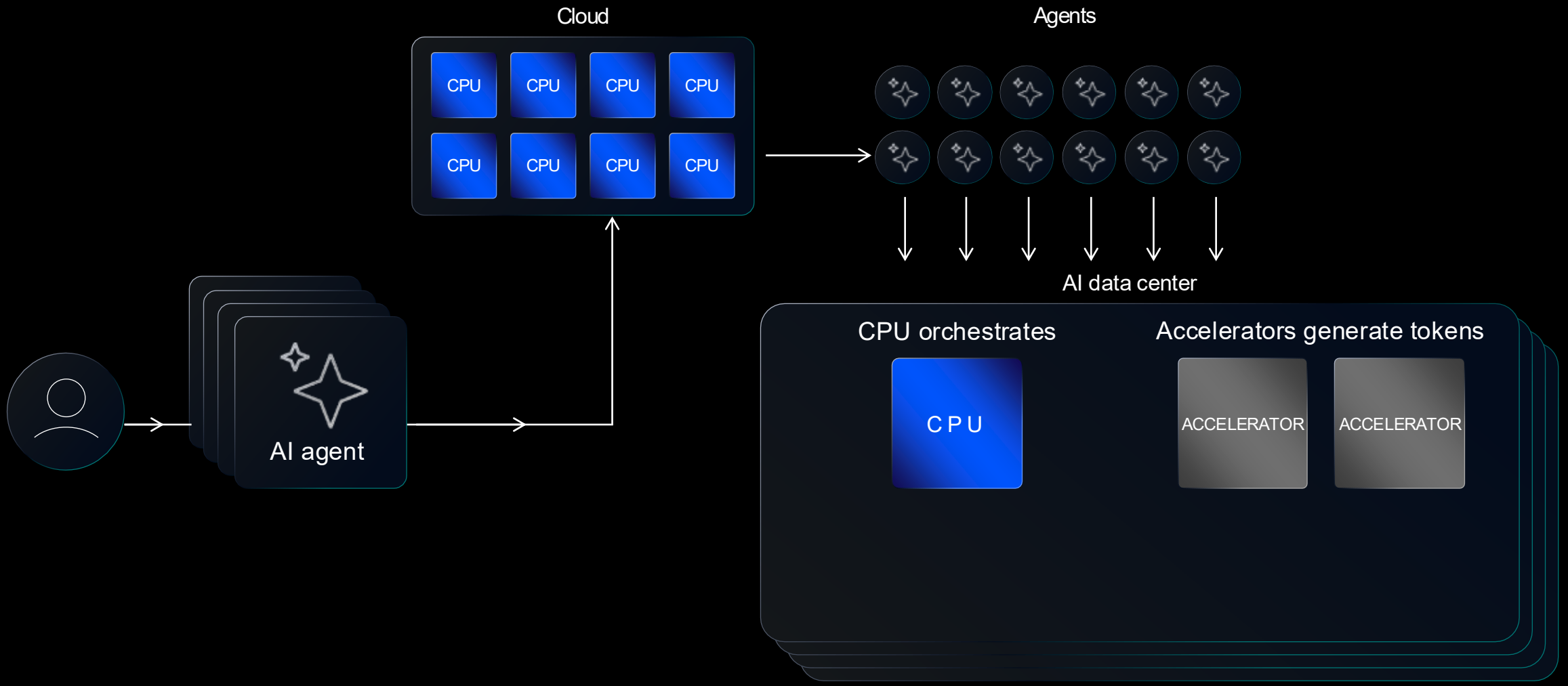


Applying OCP ORv3 and ORW Design Principles to Rack-Scale Arm AI Infrastructure with Supermicro

Daniel Chen, Supermicro

Marc Meunier, Arm

Workloads are changing...
Agents query >15x requests of humans



Start with the workload, then size the rack

A breakdown of an agentic AI workload

Workload role	Example activity	Rack design question
CPU and runtime	Orchestration, tools and services	How much memory, I/O and CPU capacity?
Data services	Retrieval, storage and data movement	Which bandwidth and locality constraints?
Model serving	Inference on the selected compute	What power and cooling envelope?

Five design choices determine whether AI infrastructure scales

DENSITY

How much useful compute can the rack support?

POWER

How is that power delivered reliably?

THERMAL

How is heat removed at sustained load?

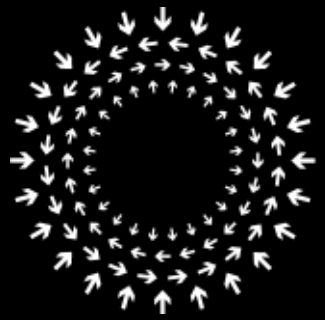
MODULARITY

Can compute and infrastructure evolve independently?

SERVICEABILITY

Can operators maintain the rack efficiently?

OCP provides the open rack framework that connects these choices.



OPEN Compute Project®

+

arm

Arm contributions across the full stack

Chiplet enablement

Server standardization

Server designs

Diverse silicon ecosystem

- Foundation Chiplet System Architecture (FCSA)
- Modular integration of compute, memory, and I/O
- Reduced design complexity for custom silicon

Software just works on Arm servers

- Standardized server architecture (SBSA, SBMR)*
- Compliance and validation framework (ADAC)
- Built-in system management and security

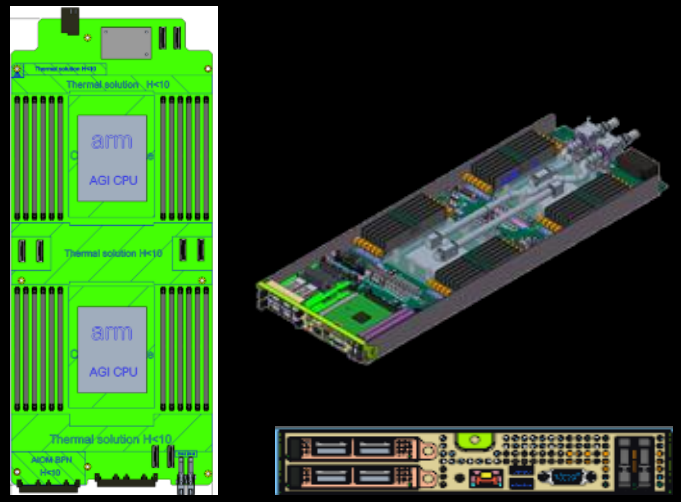
** Contributions in progress*

Helping OEMs launch Arm servers faster

- Pre-defined server hardware specifications^
- Aligned with Meta Yosemite V5 rack design
- Firmware development frameworks

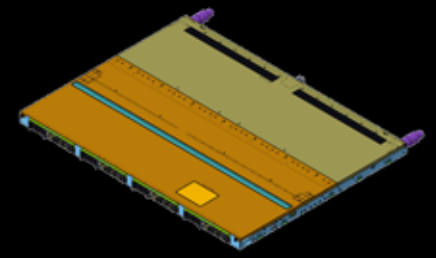
^ Future contribution

Supermicro Arm Neoverse ORv3 & ORW



Arm Dual-processor compute node

- Dual-Processor MB
- AR3DPT-64C/128C/136C
- 1x DC-SCM with I/O (Open BMC)
- 2x 400G/800G OCP-NIC
- 2x E1.S/M.2



ORW rack

42” wide 1 OU4N8S system



ORv3 rack

21” wide 2 OU4N8S system



1U Multi-Node ORW AGI server

ARS-122GT-HEOW-LCC

2U Multi-Node AGI server

ARS-222GT-HEO-LCC

ORv3 LQ Rack-Scale Agentic AI Solutions with Arm AGI CPUs

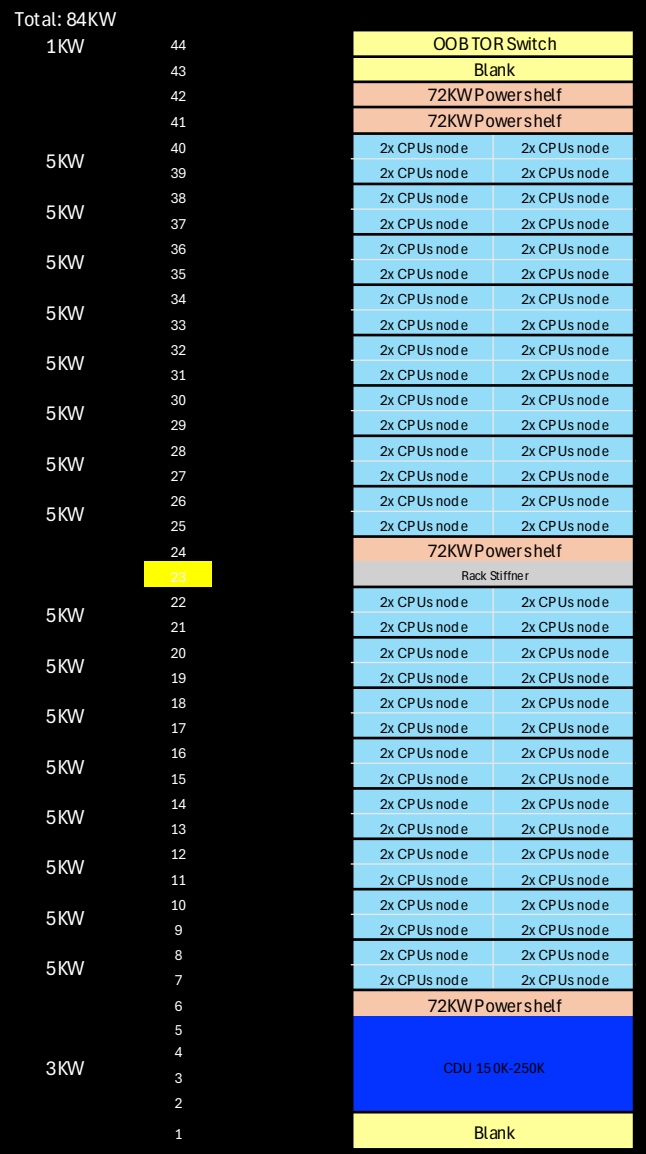
- Liquid cooled standard 21” ORV3 44OU Rack.
- Total 16 nodes 2OU4N, 128 AGI CPUs per rack.
- Total ~82.5kw per rack. Three 72 kW Power Shelves provide 2+1 redundancy at the power shelf level.
- At the PSU level, the design offers redundancy exceeding N+N requirements.
- Six 415V/60A power chips.
- One in-rack CDU.



2 OU4N ORV3
LIQUID COOLED RACK

Up to 22,848 Arm AGI CPU Cores
2x NVMe E1.S drives
2x PCIe Gen 6 slots
24x DIMM Slots

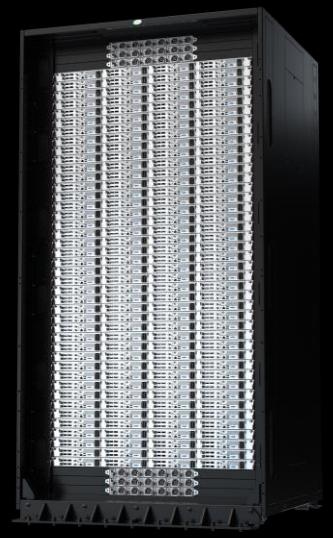
Standard 44-OU ORV3 rack
(48V Bus Bar, DLC rack)
(64 node, 128 CPUs, 84KW)



Scan to see more details on the servers and racks

ORW Rack-Scale Agentic AI Solutions with Arm AGI CPUs

- Liquid cooled standard 42” ORW 44OU Rack.
- Total 148 nodes 1OU4N, 296 AGI CPUs per rack.
- Total ~185kw per rack. Six 72 kW Power Shelves provide 4+2 redundancy at the power shelf level.
- At the PSU level, the design offers redundancy exceeding N+N requirements.
- 12 415V/60A power whips.
- One In-Row CDU.



1 OU4N ORW
LIQUID COOLED RACK

Up to 40,256 Arm AGI CPU Cores
2x NVMe E1.S drives
2x PCIe Gen 6 slots
24x DIMM Slots

Supermicro 48-OU ORV rack
(48V Bus Bar, DLC rack)
(148 nodes, 296 CPUs, 185KW)

Total: 185KW					
	48		72KW or 74KW Power shelf		
	47		72KW or 74KW Power shelf		
	46		72KW or 74KW Power shelf		
5KW	45				
5KW	44	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	43	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	42	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	41	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	40	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	39	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	38	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	37	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	36				
5KW	35	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	34	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	33	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	32	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	31	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	30	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	29	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	28	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	27	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	26				
5KW	25	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	24	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	23	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	22	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	21	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	20	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	19	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	18	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	17				
5KW	16	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	15	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	14	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	13	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	12	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	11	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	10	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	9	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	8	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	7	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	6	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	5	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
5KW	4	2x CPUs node	2x CPUs node	2x CPUs node	2x CPUs node
	3		72KW or 74KW Power shelf		
	2		72KW or 74KW Power shelf		
	1		72KW or 74KW Power shelf		

Scan to see more details on the servers and racks

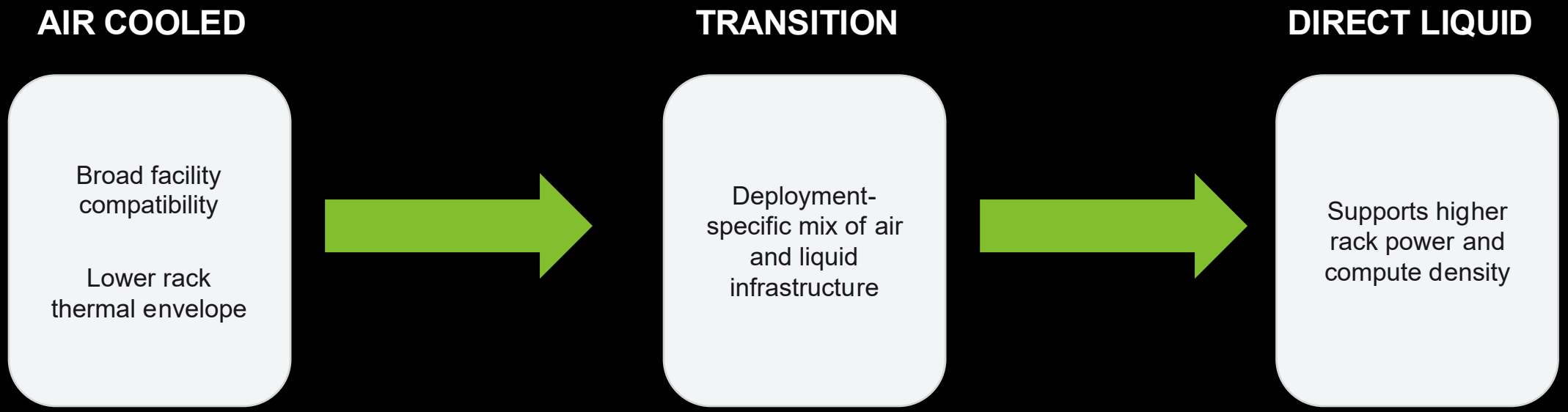
ORv3 and ORW: same design principles, different scale envelope

Design dimension	ORv3 example	ORW example
Rack architecture	21-inch / 44OU	42-inch / wider frame
AGI CPUs per rack	128	296
Rack power	~82.5 kW	~185 kW
Power shelves	3 x 72 kW	6 x 72 kW
Cooling	Direct liquid cooling	Direct liquid cooling
CDU location	In-rack/In-row	In-rack/In-Row

As compute density increases, power, cooling and facility integration scale with it.

Cooling architecture follows the deployment envelope

OCP-aligned designs can span air-cooled and liquid-cooled environments



The cooling choice is part of the rack architecture — not an afterthought.

Open interfaces reduce deployment friction

Rack standards become repeatable integration and service checks

POWER

Voltage, connectors, power budget, redundancy and fault response

COOLING

Coolant conditions, interface limits, routing and service access

MECHANICAL

Envelope, clearances, insertion path and cable routing

MANAGEMENT

Topology, ports, telemetry, control ownership and recovery

Serviceability is the operational test of the architecture

Replace compute node

Service power shelf

Service cooling connection

Access network and cabling



Three principles for scalable AI infrastructure

1
START WITH THE WORKLOAD

CPU, accelerator, memory, storage and network requirements determine the rack envelope.

2
STANDARDIZE THE BOUNDARIES

Open power, cooling, mechanical and management interfaces reduce integration complexity.

3
DESIGN THE RACK AS THE SYSTEM

Compute, power, cooling, modularity and serviceability must scale together.

OCP-aligned infrastructure creates a common framework for turning rapid compute innovation into deployable AI systems.


Thank
You!
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